

Rakumako Moloko

Junior Data Scientist | Machine learning Engineer

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SUMMARY

Junior Data Scientist and Machine Learning Engineer with hands-on experience in Python, TensorFlow, Scikit-learn, and Power BI, gained through software testing and digital transformation roles at Softelsie Solutions and Nedbank Group. Proven ability to build production-ready ML models for time-series forecasting, predictive maintenance, and anomaly detection. Strong foundation in data analytics, feature engineering, and Agile project delivery. Experienced in handling imbalanced datasets, model evaluation, and translating business problems into technical solutions.

EDUCATION

Degree in Bachelor of Computing (Data Science)

Belgium Campus ITVersity | 2023 – 2025

Currently Pursuing: **Honours In Bachelor of Computing**
(Data Science)

Belgium Campus ITVersity | 2026

National Senior Certificate (Matric)

Mathew Phosa High School | 2022

WORK EXPERIENCE

Softelsie Solutions PTY (LTD) Randburg

Software Tester (Learnership) | Jan 2025 – Mar 2025

- Executed 150+ test cases across web and mobile applications, identifying and documenting 45+ critical defects that improved product stability by 30%
- Collaborated with development team using Agile methodology to reproduce bugs, verify fixes, and ensure quality standards were met before production releases
- Created detailed test documentation and bug reports in JIRA, reducing average bug resolution time by 25%
- Performed regression, functional, and user acceptance testing on financial services applications
- Gained hands-on experience with test automation frameworks (Selenium) and API testing tools (Postman)
- Utilized SQL for database validation and Git for version control in testing workflows

Nedbank Group Technology Programme

IT Trainee - Digital Transformation Programme (Contract) | May 2025 – Aug 2025

- Completed comprehensive assessments in Machine Learning, TensorFlow, PyTorch, and advanced AI concepts
- Developed Anti-Money Laundering (AML) detection system proposal using ML to identify suspicious transactions
- Participated in enterprise-scale digital transformation initiative affecting 8+ million customers

TOP PROJECTS

Electricity Demand Forecasting (Machine Learning & Time-Series Project)

- Built time-series ML models to forecast hourly electricity demand with 95%+ accuracy using XGBoost and Random Forest
- Engineered 38+ features including lag variables, rolling statistics, and temporal patterns from 8,760+ hours of consumption data
- Implemented LSTM neural network using TensorFlow for deep learning comparison with classical ML approaches
- Performed comprehensive model evaluation using MAE, RMSE, and MAPE metrics to ensure production-ready performance
- Developed complete data processing pipeline including missing value handling, outlier removal, and feature scaling
- Business Value: Enables optimized generation scheduling and capacity planning for power grid operators

Fault Detection Using Deep Learning (Unsupervised Anomaly Detection Project)

- Implemented unsupervised anomaly detection system using autoencoder neural networks achieving 96% accuracy
- Designed and trained deep autoencoder architecture using TensorFlow/Keras
- Achieved perfect recall (100%) on anomalies with clear separation between normal and anomalous patterns
- Implemented automated threshold determination using 95th percentile of training reconstruction errors
- Business Value: Enables early identification of abnormal patterns without requiring labeled failure data

Health and Demographic Patterns in South Africa (Academic Data Science Project)

- Analyzed demographic and health data from South African population surveys using statistical modeling in R
- Built and compared multiple classification models (Logistic Regression, Decision Trees, Random Forest, Naive Bayes)
- Created comprehensive data visualizations and statistical reports for stakeholder presentation
- Demonstrated end-to-end data science workflow from data understanding to model deployment

ADDITIONAL COURSES AND ACHIEVEMENTS

1. ALX Africa - AI Career Essentials

2. Kaggle Certifications:

- Intermediate Machine Learning
- Deep Learning Specialization
- Python Programming

3. Driver's License (Code 10)

SKILLS

- Machine Learning & AI: Scikit-learn, XGBoost, Random Forest, TensorFlow, Keras, PyTorch, Deep Learning, Neural Networks, CNNs, LSTMs, Autoencoders, Time-series Forecasting, Classification, Anomaly Detection, Feature Engineering, Model Evaluation, SMOTE, SHAP
- Data Science & Analytics: Python, Pandas, NumPy, R, SQL, Statistical Analysis, Data Mining, EDA, Data Visualization (Matplotlib, Seaborn), Data Preprocessing
- Programming & Development: Python (Advanced), R (Intermediate), C# (Intermediate), SQL (Intermediate), OOP, Git, GitHub
- Software Testing & QA: JIRA, Selenium, Postman, Functional Testing, Regression Testing, UAT, API Testing
- Cloud & BI: Microsoft Azure (Exposure), Power BI, DAX, Tableau (Basic), Excel (Advanced)
- FinTech & Domain: Digital Banking, AML, Credit Risk, Financial Regulations, Ethical AI
- Methodologies: Agile/Scrum, CRISP-DM

TECHNICAL STACK

- Languages: Python, R, C#, SQL
- ML Libraries: Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Pandas, NumPy
- Data Viz: Matplotlib, Seaborn, Power BI
- Databases: SQL Server, MySQL
- Cloud: Azure (Exposure)
- Testing Tools: JIRA, Selenium, Postman
- Version Control: Git, GitHub
- IDEs: Jupyter Notebook, VS Code, RStudio, PyCharm
- BI Tools: Power BI
- APIs: Flask REST APIs

LEADERSHIP AND ACTIVITIES

- Res Rep- Resident Representative Council (RRC)
- Snooker / Pool
- Chess
- Soccer